

CO3 AT2_Medium Level Class Test

View Only

QUESTIONS

Q1

Default and Parameterized Constructors
20 marks

Q2

Constructor Overloading
20 marks

Q3

Copy Constructor with Dynamic Memory
20 marks

Q4

Destructor and Resource Management
20 marks

Q5

Overloading the + Operator (Complex Numbers)
20 marks

Q1 Default and Parameterized Constructors

Write a C++ program that defines a class Book with attributes title, author, and price. Implement:

- A. A default constructor that initializes all attributes to default values ("Unknown", "Unknown", 0.0).
- B. A parameterized constructor that accepts values for all three attributes.
- C. A display() member function to print the book details.

In main(), create one object using each constructor and display its details.

Your Code

```
1 #include<iostream>
2 #include<string>
3 using namespace std;
4 class Book{
5     string title,author;
6     double price;
7     public:
8     Book()
9     {
10         title="unknown";
11         author="unknown";
12         price=0.0;
```

Input

stdin in

Output

Search



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Q2 Constructor Overloading

Write a C++ program that defines a class Rectangle with the following constructors:

- A. A default constructor that sets length and breadth to 1.
- B. A single-parameter constructor that creates a square (length = breadth).
- C. A two-parameter constructor that sets both length and breadth.

Include a member function area() to compute the rectangle's area. In main(), create three objects and compute their areas.

Your Code

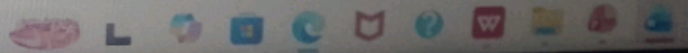
```
1 #include<iostream>
2 using namespace std;
3 class Rectangle
4 {
5     double length,breadth;
6     public:
7     Rectangle()
8     {
9         length=1;
10        breadth=1;
11    }
```

28°C

light rain



Search



Q2

Constructor Overloading
20 marks

Q3

Copy Constructor with Dynamic
Memory
20 marks

Q4

Destructor and Resource
Management
20 marks

Q5

Overloading the + Operator
(Complex Numbers)
20 marks

5/5 answered

- A. A parameterized constructor that allocates memory and copies the name.
- B. A deep copy constructor that allocates new memory for the copied object.
- C. A destructor that releases the allocated memory.
- D. A display() function to print the student's details.

In main(), create a Student object, create another object using the copy constructor, and modify the original remains unaffected.

Your Code

```

1 #include<iostream>
2 #include<cstring>
3 using namespace std;
4 class Student{
5     char *name;
6     int marks;
7     public:
8     student(const char*n,int m){
9         name=new char[strlen(n)+1];
10        strcpy(name,n);
11        marks=m;
12    }
13    student(const student &s){
14        name=new char[strlen(s.name)+1];
15        strcpy(name,s.name);
16        marks=s.marks;
17    }
18    ~ student(){
19        delete[]name;

```


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Q4 Destructor and Resource Management

Write a C++ program that defines a class FileHandler which simulates opening and closing a file.

A. A constructor that prints "File <name> opened".

B. A destructor that prints "File <name> closed".

In main(), create three FileHandler objects inside a block { } and observe the order in which the destructor message is printed after the block ends to confirm the destruction order.

Your Code

```
1 #include<iostream>
2 #include<string>
3 using namespace std;
4 class Filehandler
5 {
6     string name;
7     public:
8     Filehandler(string n)
9     {
10         name=n;
11         cout<<"file"<<name<<"opened"<<endl;
12     }
```

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20 marks



Q5 Overloading the + Operator (Complex Numbers)

Write a C++ program that defines a class Complex with real and imaginary parts. Overload and return a new Complex object. Include a display() function to print the result in the form. In main(), create two Complex objects, add them using the overloaded + operator, and display the result.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class Complex {
4     int real, imag;
5 public:
6     Complex(int r = 0, int i = 0) {
7         real = r;
8         imag = i;
9     }
10    Complex operator+(Complex c) {
11        Complex temp;
12        temp.real = real + c.real;
13        temp.imag = imag + c.imag;
14        return temp;
15    }
```